The Bootstrap

Last modified: 2026-10-15 14:18:11 (PDT)

This page reviews the bootstrap, a resampling method for standard errors and confidence intervals that does not need a formula for the sampling distribution of a statistic. Its HERS example bootstraps the slope of a simple linear regression. This page is adapted from Vittinghoff et al. (2012), Section 3.6.

The HERS data

The examples on this page use the HERS data, which the Comparing Means page describes. The `rmb` R package includes the dataset; `haven::as_factor()` converts its Stata value labels to factors:

```
hers <- rmb::hers |> haven::as_factor()
hers |> head()
#> # A tibble: 6 x 37
#>   HT      age raceth nonwhite smoking drinkany exercise physact globrat poorfair
#>   <fct> <dbl> <fct>  <fct>    <fct>   <fct>    <fct>    <fct>   <fct>   <fct>
#> 1 plac~    70 Afric~ yes      no      no       no       much m~ good    no
#> 2 plac~    62 Afric~ yes      no      no       no       much l~ good    no
#> 3 horm~    69 White  no       no      no       no       about ~ good    no
#> 4 plac~    64 White  no       yes     yes      no       much l~ good    no
#> 5 plac~    65 White  no       no      no       no       somewh~ good    no
#> 6 horm~    68 Afric~ yes      no      yes      no       about ~ good    no
#> # i 27 more variables: medcond <dbl>, htnmeds <fct>, statins <fct>,
#> #   diabetes <fct>, dmpills <fct>, insulin <fct>, weight <dbl>, BMI <dbl>,
#> #   waist <dbl>, WHR <dbl>, glucose <dbl>, weight1 <dbl>, BMI1 <dbl>,
#> #   waist1 <dbl>, WHR1 <dbl>, glucose1 <dbl>, tchol <dbl>, LDL <dbl>,
#> #   HDL <dbl>, TG <dbl>, tchol1 <dbl>, LDL1 <dbl>, HDL1 <dbl>, TG1 <dbl>,
#> #   SBP <dbl>, DBP <dbl>, age10 <dbl>
```

Bootstrap confidence intervals

When to use the bootstrap

The bootstrap (Efron 1979; Efron and Tibshirani 1993) estimates the sampling distribution of a statistic by resampling the observed data. It gives standard errors and confidence intervals in three situations where the usual formulas fall short (Vittinghoff et al. 2012, chap. 3):

- approximate methods for valid confidence intervals exist, but standard software does not implement them conveniently;
- no closed-form approximate method has been found;
- the data violate the assumptions of the established methods badly enough that their confidence intervals would be unreliable.

The bootstrap procedure

Definition 0.1 (Bootstrap sample)**.** A **bootstrap sample** from observed data $x_1, \ldots, x_n$ is a sample of size n drawn with replacement from $\{x_1, \ldots, x_n\}$, each draw choosing each of the n observations with probability $1/n$.

Remark (Repeated observations in a bootstrap sample)

Remark 0.1 (Repeated observations in a bootstrap sample). Because the draws are with replacement, a bootstrap sample usually contains some observations more than once and omits others. For example, a bootstrap sample from the data $1, 2, 3$ might be $2, 2, 3$.

Example 0.1 (Bootstrap samples of five glucose values)**.** Take the baseline fasting glucose values of the first five HERS participants, and draw two bootstrap samples from them:

```r
glucose5 <- hers$glucose[1:5]
glucose5
#> [1]  84 111 114  94 101
set.seed(1)
list(
  sample_1 = sample(glucose5, replace = TRUE),
  sample_2 = sample(glucose5, replace = TRUE)
)
#> $sample_1
#> [1]  84  94  84 111 101
#>
#> $sample_2
#> [1] 114 111 114 114  84
```

Each bootstrap sample has five values, but some original values appear more than once and others do not appear at all.

Definition 0.2 (Bootstrap distribution)**.** Let $\hat{\theta}$ be a statistic computed from the observed data. Draw B independent bootstrap samples, and compute the statistic on each one, giving $\hat{\theta}^*_1, \ldots, \hat{\theta}^*_B$. The **bootstrap distribution** of $\hat{\theta}$ is the empirical distribution of $\hat{\theta}^*_1, \ldots, \hat{\theta}^*_B$.

Remark (What the bootstrap distribution estimates)

Remark 0.2 (What the bootstrap distribution estimates)*.* The bootstrap distribution estimates the sampling distribution of $\hat{\theta}$. The observed sample stands in for the population, and resampling from it stands in for drawing new samples from the population.

Definition 0.3 (Bootstrap standard error)**.** The **bootstrap standard error** of a statistic $\hat{\theta}$, written $\widehat{\mathrm{SE}}_{\mathrm{boot}}$, is the sample standard deviation of the bootstrap replicates $\hat{\theta}^*_1, \ldots, \hat{\theta}^*_B$ (Definition 0.2). It estimates the standard error of $\hat{\theta}$.

Example 0.2 (Bootstrap distribution of a mean of ten glucose values)**.** Take the baseline fasting glucose values of the first ten HERS participants, and compute the sample mean of each of $B = 1{,}000$ bootstrap samples:

```r
glucose10 <- hers$glucose[1:10]
set.seed(42)
boot_means10 <- replicate(1000, mean(sample(glucose10, replace = TRUE)))
c(
  observed_mean = mean(glucose10),
  bootstrap_se = sd(boot_means10),
  formula_se = sd(glucose10) / sqrt(10),
  divide_by_n_se = sqrt(mean((glucose10 - mean(glucose10))^2) / 10)
)
#>  observed_mean   bootstrap_se     formula_se divide_by_n_se
#>      103.80000        3.47992        3.61417        3.42870
```

The bootstrap standard error (Definition 0.3) is close to the usual estimate $s/\sqrt{n}$. Each bootstrap draw has variance $\hat{\sigma}^2 \stackrel{\text{def}}{=} \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$ around $\bar{x}$, and the n draws are independent, so as B grows the bootstrap standard error of the mean approaches $\hat{\sigma}/\sqrt{n}$ (`divide_by_n_se`), which is smaller than $s/\sqrt{n}$ by the factor $\sqrt{(n-1)/n}$. Figure 1 shows the bootstrap distribution.

```r
tibble::tibble(mean = boot_means10) |>
  ggplot2::ggplot() +
  ggplot2::aes(x = mean) +
  ggplot2::geom_histogram(bins = 30, color = "white") +
  ggplot2::geom_vline(xintercept = mean(glucose10), color = "red") +
  ggplot2::labs(x = "Bootstrap mean (mg/dL)", y = "Count")
```

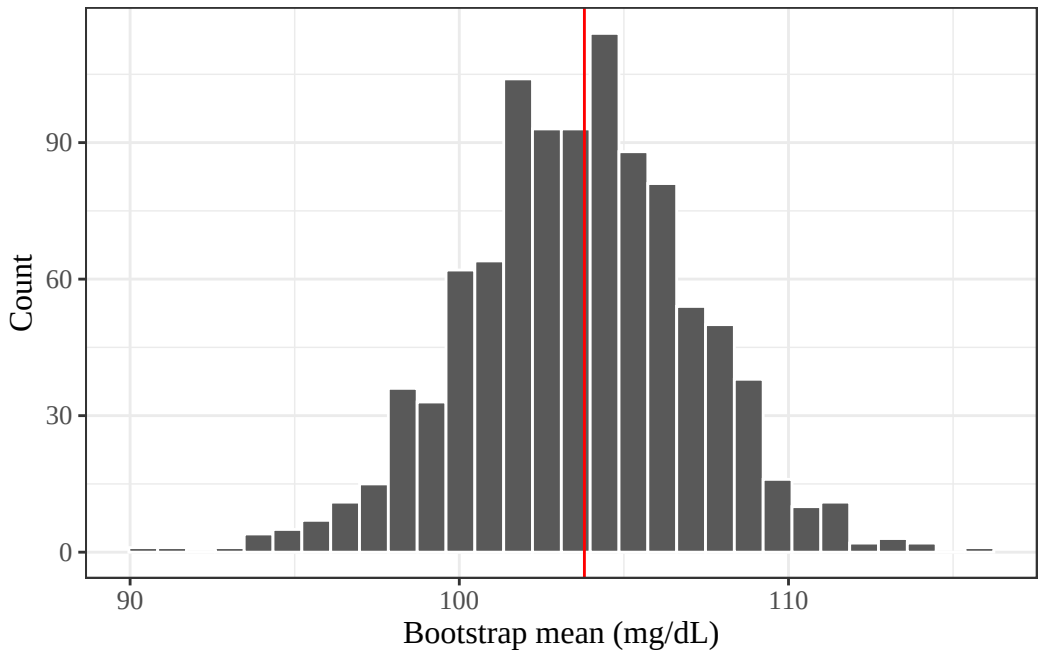


Figure 1: Bootstrap distribution of the mean fasting glucose of the first ten HERS participants ($B = 1{,}000$). The red line marks the observed mean.

Read

- Hogg et al. (2019, sec. 7.7, pp. 353–360) covers resampling methods.
- Casella and Berger (2002, sec. 10.1.4, pp. 478–480) covers bootstrap standard errors.
- Rice (2007, sec. 10.4.6, pp. 399–400) covers variability of location estimates by the bootstrap.
- Rice (2007, sec. 14.6, pp. 585–586) covers conditional inference, unconditional inference and the bootstrap.

Bootstrap confidence interval methods

There are three common methods for turning a bootstrap distribution into a $100(1-\alpha)\%$ confidence interval for a parameter θ (Vittinghoff et al. 2012, chap. 3; Efron and Tibshirani 1993). Each is an approximate confidence interval. Throughout, z_q is the q quantile of the standard Gaussian distribution and Φ is its CDF.

Normal approximation

Definition 0.4 (Normal bootstrap confidence interval)**.** The **normal bootstrap confidence interval** is

$$\hat{\theta} \pm z_{1-\alpha/2}\, \widehat{\mathrm{SE}}_{\mathrm{boot}},$$

where $\widehat{\mathrm{SE}}_{\mathrm{boot}}$ is the bootstrap standard error (Definition 0.3).

Remark (When the normal bootstrap interval is reliable)

Remark 0.3 (When the normal bootstrap interval is reliable). This interval assumes that the sampling distribution of $\hat{\theta}$ is approximately Gaussian and centered at θ, so it can be unreliable when that distribution is skewed. It needs only a standard error, which takes fewer bootstrap replicates to estimate well than the tail quantiles used by the percentile and BCa intervals (Efron and Tibshirani 1993).

Example 0.3 (Normal bootstrap interval for the mean of ten glucose values)**.** Continuing Example 0.2, the 95% interval of Definition 0.4 is:

```
mean(glucose10) +
  c(lower = -1, upper = 1) * qnorm(0.975) * sd(boot_means10)
#>    lower    upper
#>  96.9795 110.6205
```

Percentile method

Definition 0.5 (Percentile bootstrap confidence interval)**.** The **percentile bootstrap confidence interval** runs from the $\alpha/2$ quantile to the $1-\alpha/2$ quantile of the bootstrap replicates $\hat{\theta}^*_1, \ldots, \hat{\theta}^*_B$ (Definition 0.2).

Remark (Number of bootstrap replicates for percentile intervals)

Remark 0.4 (Number of bootstrap replicates for percentile intervals). The extreme quantiles of B replicates are noisy estimates, so percentile-based intervals need more replicates than the normal interval; B of at least 1,000 is a common choice.

Example 0.4 (Percentile bootstrap interval for the mean of ten glucose values)**.** Continuing Example 0.2, the 95% interval of Definition 0.5 is:

```
quantile(boot_means10, c(0.025, 0.975))
#>  2.5% 97.5%
#>  96.5 110.4
```

Bias-corrected and accelerated method

Definition 0.6 (Bias correction of the BCa interval)**.** For a statistic $\hat{\theta}$ and its bootstrap replicates $\hat{\theta}^*_1, \dots, \hat{\theta}^*_B$, the **bias correction** is

$$\hat{z}_0 \stackrel{\text{def}}{=} \Phi^{-1}\left(\frac{\#\left\{b : \hat{\theta}^*_b < \hat{\theta}\right\}}{B}\right),$$

where Φ is the standard Gaussian CDF.

Example 0.5 (Bias correction when 40% of replicates fall below the estimate)**.** If 400 of $B = 1{,}000$ replicates are below $\hat{\theta}$, then $\hat{z}_0 = \Phi^{-1}(0.4) \approx -0.253$. If exactly half were below,

$$\begin{aligned}\hat{z}_0 &= \Phi^{-1}(0.5)\\ &= 0.\end{aligned}$$

Definition 0.7 (Acceleration of the BCa interval)**.** Let $\hat{\theta}_{(i)}$ be the statistic computed with observation i deleted, and let $\hat{\theta}_{(\cdot)}$ be the mean of $\hat{\theta}_{(1)}, \dots, \hat{\theta}_{(n)}$. The **acceleration** is

$$\hat{a} \stackrel{\text{def}}{=} \frac{\sum_{i=1}^{n}\left(\hat{\theta}_{(\cdot)} - \hat{\theta}_{(i)}\right)^3}{6\left(\sum_{i=1}^{n}\left(\hat{\theta}_{(\cdot)} - \hat{\theta}_{(i)}\right)^2\right)^{3/2}}.$$

Example 0.6 (Acceleration from three deleted estimates)**.** If the differences $\hat{\theta}_{(\cdot)} - \hat{\theta}_{(i)}$ are 1, 1, and -2, the sum of their cubes is $1+1-8=-6$ and the sum of their squares is 6, so

$$\hat{a} = \frac{-6}{6 \cdot 6^{3/2}} \approx -0.068.$$

Definition 0.8 (Bias-corrected and accelerated bootstrap confidence interval)**.** With the bias correction $\hat{z}_0$ and the acceleration $\hat{a}$, for $q \in \{\alpha/2, 1-\alpha/2\}$, let

$$\alpha_q \overset{\text{def}}{=} \Phi\left(\hat{z}_0 + \frac{\hat{z}_0 + z_q}{1 - \hat{a}(\hat{z}_0 + z_q)}\right).$$

The **bias-corrected and accelerated (BCa) bootstrap confidence interval** runs from the $\alpha_{\alpha/2}$ quantile to the $\alpha_{1-\alpha/2}$ quantile of the bootstrap replicates (Efron and Tibshirani 1993, chap. 14).

Remark (What the BCa interval corrects)

Remark 0.5 (What the BCa interval corrects). When $\hat{z}_0 = 0$ and $\hat{a} = 0$, $\alpha_q = q$ and the BCa interval is the percentile interval (Definition 0.5). $\hat{z}_0$ measures how far the bootstrap distribution's median sits from $\hat{\theta}$, and $\hat{a}$ measures its skewness, so the BCa interval shifts the percentile interval to correct for both.

Example 0.7 (BCa bootstrap interval for the mean of ten glucose values)**.** Continuing Example 0.2, the quantities of Definition 0.8, step by step:

```
z0 <- qnorm(mean(boot_means10 < mean(glucose10)))
jackknife_means <- vapply(1:10, \(i) mean(glucose10[-i]), numeric(1))
dev <- mean(jackknife_means) - jackknife_means
a_hat <- sum(dev^3) / (6 * sum(dev^2)^(3 / 2))
z_q <- qnorm(c(0.025, 0.975))
alpha_q <- pnorm(z0 + (z0 + z_q) / (1 - a_hat * (z0 + z_q)))
c(z_0 = z0, a_hat = a_hat, alpha_lower = alpha_q[1], alpha_upper = alpha_q[2])
#>         z_0       a_hat alpha_lower alpha_upper
#>  0.01002668 -0.00867722  0.02422093  0.97422713
quantile(boot_means10, alpha_q)
#> 2.422093% 97.42271%
#>      96.5     110.4
```

The acceleration $\hat{a}$ is nearly 0 here, and $\hat{z}_0$ is small, so the BCa interval is close to the percentile interval of Example 0.4.

The boot package

The `boot` package (Davison and Hinkley 1997), a recommended package distributed with R, provides `boot::boot()` to draw the bootstrap replicates and `boot::boot.ci()` to compute the three intervals. `boot::boot.ci()` differs from the definitions on this page in two small ways:

- its normal interval (`type = "norm"`) is centered at $\hat{\theta}$ minus the bootstrap estimate of bias, $\hat{\theta} - (\bar{\theta}^* - \hat{\theta})$, where $\bar{\theta}^*$ is the mean of the replicates, rather than at $\hat{\theta}$;
- it estimates quantiles of the replicates by interpolation, so its percentile and BCa endpoints can differ slightly from those computed with `quantile()`.

Example: slope of SBP on age in HERS

Example 0.8 (Bootstrap confidence intervals for the slope of SBP on age in HERS)**.** We regress systolic blood pressure (`SBP`) on `age` by ordinary least squares, and bootstrap the slope, resampling participants (adapted from Vittinghoff et al. 2012, chap. 3). The statistic function takes the data and the row indices of one bootstrap sample:

```
slope_sbp_age <- function(data, indices) {
  fit <- lm(SBP ~ age, data = data[indices, ])
  coef(fit)[["age"]]
}

set.seed(42)
boot_result <- boot::boot(data = hers, statistic = slope_sbp_age, R = 1000)
boot_result
#>
#> ORDINARY NONPARAMETRIC BOOTSTRAP
#>
#>
#> Call:
#> boot::boot(data = hers, statistic = slope_sbp_age, R = 1000)
#>
#>
#> Bootstrap Statistics :
#>     original       bias    std. error
#> t1* 0.471728 -0.000635537   0.0536514
```

The bootstrap standard error is close to the model-based standard error from `lm()`, and the three bootstrap intervals are close to the model-based 95% confidence interval:

```r
fit_sbp_age <- lm(SBP ~ age, data = hers)
parameters::parameters(fit_sbp_age) |>
  parameters::print_md(digits = 3, ci_digits = 3)
```

Parameter	Coefficient	SE	95% CI	t(2761)	p
(Intercept)	103.629	3.596	(96.579, 110.680)	28.820	< .001
age	0.472	0.054	(0.366, 0.577)	8.787	< .001

```r
boot::boot.ci(boot_result, type = c("norm", "perc", "bca"))
#> BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
#> Based on 1000 bootstrap replicates
#>
#> CALL :
#> boot::boot.ci(boot.out = boot_result, type = c("norm", "perc",
#>     "bca"))
#>
#> Intervals :
#> Level      Normal             Percentile            BCa
#> 95%   ( 0.3672,  0.5775 )   ( 0.3607,  0.5811 )   ( 0.3624,  0.5824 )
#> Calculations and Intervals on Original Scale
```

All three intervals are similar here. When the bootstrap distribution is skewed, the intervals differ more, and the BCa interval, which corrects for skewness, is the better choice (Efron and Tibshirani 1993, chap. 14).

References

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Davison, Anthony C., and David V. Hinkley. 1997. *Bootstrap Methods and Their Application*. Cambridge Series in Statistical and Probabilistic Mathematics 1. Cambridge University Press. https://doi.org/10.1017/CBO9780511802843.

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